

## FRAMEWORK MANUFACTURING SERVICES AGREEMENT

## MSA-CAL-AST-2024-011

Effective 18 November 2024 between Caldova Pharmaceuticals, Inc. and Aster Ridge Biomanufacturing.

**Purpose.** This agreement establishes commercial, operational, quality, and governance terms under which Caldova may issue product schedules and purchase orders for pharmaceutical manufacturing services.

## 1. Parties and term

**1.1 Parties.** Caldova Pharmaceuticals, Inc. (“Caldova”) and Aster Ridge Biomanufacturing, with its manufacturing site in Research Triangle Park, North Carolina, USA (“Manufacturer”), enter this Agreement as independent contractors.

**1.2 Term.** The initial term is Five years from the Effective Date unless terminated earlier. Product schedules, amendments, quality agreements, and accepted purchase orders are incorporated by reference.

**1.3 Order of precedence.** The applicable quality agreement controls GMP responsibilities; then a signed amendment or product schedule; then this Agreement; then an accepted purchase order. Supplier acknowledgments do not add conflicting terms.

## 2. Governance and planning

**2.1 Joint oversight.** A joint operating committee reviews forecasts, capacity, quality, changes, performance, and continuity at least quarterly.

**2.2 Forecasts.** Rolling 12-month forecast updated monthly; first three months binding after product-specific award

**2.3 Purchase orders.** Each order identifies product, batch count or quantity, delivery dates, destination, storage and shipping conditions, and required release documents. Manufacturer shall promptly accept or identify a conflict.

Framework ID  
MSA-CAL-AST-2024-011

Supplier master ID  
sup-001

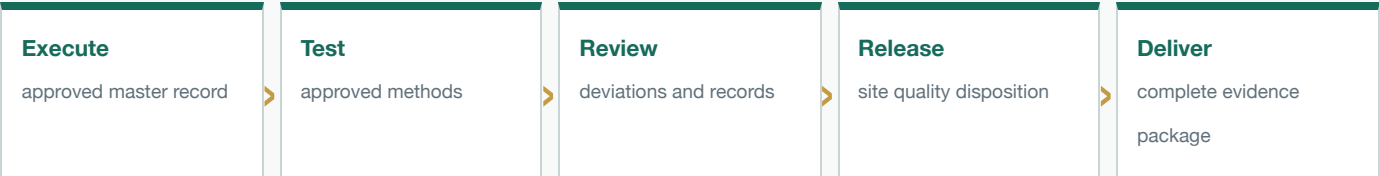
3. Manufacture and materials

- 3.1 Performance.** Manufacturer shall use qualified facilities, equipment, systems, materials, and trained personnel to manufacture each scheduled product in accordance with cGMP, approved specifications, master records, applicable law, and written Caldova instructions.
- 3.2 Materials.** Manufacturer shall qualify and control purchased materials, maintain traceable inventory, use approved sources, and notify Caldova of shortages or suspected defects. Caldova-supplied materials remain Caldova property.
- 3.3 Site and subcontractors.** Manufacture is limited to the approved Research Triangle Park, North Carolina, USA site. No GMP manufacturing, packaging, or release testing may be subcontracted without Caldova's prior written approval.
- 3.4 Changes.** Supplier must notify Caldova before changes to site, process, equipment train, critical material, test method, or approved subcontractor.

4. Testing, release, and delivery

- 4.1 Batch package.** Each batch requires a certificate of analysis, certificate of compliance, executed record package, and supplier quality-unit disposition.
- 4.2 Delivery.** Manufacturer shall release and make product available by the delivery date in the applicable order, packaged and labeled as specified. Title and risk transfer under the stated Incoterm.
- 4.3 Acceptance and rejection.** Caldova may quarantine and reject nonconforming product; supplier-caused failures are replaced without a second manufacturing charge.

BATCH EVIDENCE FLOW



- 4.4 Failure.** Manufacturer shall investigate failures, preserve evidence, notify Caldova promptly, and bear direct replacement manufacturing cost where its breach caused nonconformance.

## 5. Quality responsibilities

**5.1 Quality agreement.** A separate executed quality agreement controls GMP responsibilities if it conflicts with this commercial agreement.

| System               | Caldova responsibility                             | Manufacturer responsibility                              |
|----------------------|----------------------------------------------------|----------------------------------------------------------|
| Quality unit         | Product requirements and final Caldova disposition | Independent site disposition and compliance oversight    |
| Facilities/equipment | Approve product-specific fit                       | Qualification, maintenance, calibration, cleaning        |
| Materials            | Approve critical sources/specifications            | Supplier qualification, receipt, testing, status control |
| Laboratory           | Approve methods and specifications                 | Method transfer, testing, data integrity, OOS control    |
| Documentation        | Timely review and decisions                        | Contemporaneous records and complete batch package       |
| Change control       | Approve product-impacting changes                  | Assess and notify before implementation                  |

**5.2 Audits.** Caldova may conduct one routine quality audit per year and additional for-cause audits.

**5.3 Inspections.** Manufacturer shall notify Caldova promptly of regulatory inspections, observations, enforcement, or commitments that may affect a Caldova product or the approved site and allow reasonable review of relevant responses.

**5.4 Records.** GMP records retained for the product lifecycle plus one year, and never less than seven years.

**5.5 Complaints and recalls.** The parties shall cooperate on complaints, investigations, field actions, and recalls. Cost follows documented responsibility for the event.

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## 6. Charges and payment

**6.1 Charges.** Product-specific prices, transfer fees, capacity commitments, and permitted adjustments appear only in signed schedules or amendments. Manufacturer may invoice released batches and achieved milestones supported by required evidence.

**6.2 Payment.** Net 60 days for undisputed invoices Caldova may withhold disputed amounts while paying undisputed amounts.

**6.3 Records and audit.** Manufacturer shall retain records supporting invoiced amounts for at least seven years and provide reasonable access for verification.

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## 7. Risk allocation

**7.1 Warranties.** Each party has authority to enter the Agreement. Manufacturer warrants services will conform to applicable law, cGMP, approved requirements, and professional standards.

**7.2 Insurance.** Commercial general, product liability, workers compensation, and cyber coverage maintained at commercially reasonable levels.

**7.3 Business continuity.** Manufacturer shall maintain and periodically test continuity plans addressing utilities, labor, materials, equipment, data, logistics, and critical suppliers; material disruptions require prompt notice and recovery estimates.

**7.4 Indemnity and liability.** Each party is responsible for direct third-party claims caused by its negligence, willful misconduct, legal violation, or material breach. Neither party is liable for indirect or punitive damages except where prohibited by law or for confidentiality, fraud, or willful misconduct.

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## 8. Information and intellectual property

**8.1 Confidentiality.** Each party protects the other's confidential information and uses it only to perform the agreement.

**8.2 Ownership.** Each party retains background intellectual property. Caldova owns product-specific specifications, records, and know-how it supplies; Manufacturer retains general platform know-how. Product-specific improvements are documented in the applicable schedule.

## 9. Termination and transition

**9.1 Cause.** Either party may terminate for uncured material breach; Caldova may require reasonable transition assistance. Immediate suspension may occur for patient risk, critical quality failure, sanctions, fraud, or loss of required authorization.

**9.2 Effect.** Manufacturer shall protect materials and product, complete or stop work as directed, return Caldova property and records, and cooperate in an orderly transfer. Accrued payment and confidentiality obligations survive.

## 10. General terms

**10.1 Law and disputes.** This Agreement is governed by Delaware, USA law. Operational disputes escalate through the joint committee and then executive sponsors before formal proceedings.

**10.2 Compliance.** Each party shall comply with anti-bribery, sanctions, labor, environmental, data-protection, and trade laws. Conflicts of interest must be disclosed.

**10.3 Assignment and amendment.** Neither party may assign this Agreement without consent except to a successor in a permitted corporate transaction. Amendments must be written and signed by authorized representatives.

**10.4 Entire agreement.** This Agreement and incorporated records are the complete framework understanding for their subject matter. Invalid provisions are replaced without invalidating the remainder. Counterparts and approved electronic signatures are effective.

**For Caldova Pharmaceuticals, Inc.**  
Charlotte Waltson · VP of Procurement  
Signed 18 November 2024

**For Aster Ridge Biomanufacturing**  
Alex Mercer · Chief Commercial Officer  
Signed 18 November 2024

All entities, names, terms, and signature blocks in this agreement are fictional demonstration data and are not legal advice.